# Kaicheng Xue

ybg4wb@virginia.edu

## EDUCATION

**University of Virginia**, Charlottesville, VA — Aug 2025 – Dec 2026 (expected)
M.S. in Computer Science — GPA: 4.0/4.0
Thesis: *Markov Dynamics of Hidden States and LLM Reasoning Accuracy* (Advisor: Yangfeng Ji)

**University of Wisconsin–Madison**, Madison, WI — Sept 2021 – May 2025
B.S. in Mathematics — GPA: 3.81/4.0
B.S. in Computer Science — GPA: 3.78/4.0
Honors: College of Letters & Science Dean's List

## RESEARCH EXPERIENCE

**Graduate Research, University of Virginia** — Jan 2026 – Present
*Markov Dynamics of Hidden States in LLM Reasoning*
Advisor: Yangfeng Ji

- Proposed a method to model layer-wise hidden-state trajectories of LLMs over chain-of-thought reasoning steps as Markov chains, summarized by entropy rate and spectral gap
- Evaluated 11 models from Llama and Qwen families (base, post-trained, LoRA-fine-tuned) across four reasoning benchmarks (HotpotQA, ProofWriter, GSM8K, HSH-Math)
- Designed a rigorous evaluation protocol using nested leave-one-out cross-validation with in-fold feature selection and selection-aware permutation tests
- Found that hidden-state Markov dynamics are confounded with model size/family across different models, but reliably track accuracy changes from fine-tuning within a fixed base model

**Graduate Research, University of Virginia** — Jan 2026 – Present
*Privacy Filter Evaluation on Fragmented Information*
Advisor: Tianhao Wang

- Constructed a controlled benchmark (4,320 samples) with continuous, fragmented, and noise-injected private information, plus public-information scenarios for measuring over-filtering
- Evaluated three released privacy filters (OpenAI, Liquid AI, Meddies) and found that fragmentation lowers F1 by 0.14–0.38 for two filters, with public-information over-filtering rates of 12–88%
- Investigated whether missed fragments can be recovered without additional training; found that lowering the output-probability threshold masks 24–54% of missed fragments (AUROC 0.95–1.00 against ordinary tokens), but the signal does not separate private from public entities
- Paper submitted to WSDM 2027

**Student Researcher, University of Virginia** — Aug 2026 – Present
*LLM-Generated Fake-Review Attacks on AI Shopping Agents*
PI: Tianhao Wang (CS); Co-PIs: Michael Albert (Darden), Jingjing Li (McIntire)

- Built an end-to-end simulation of an Amazon-style marketplace in which ∼200 LLM shopper agents browse, compare, and purchase products per target, to measure how coordinated fake-review attacks shift purchasing behavior at the population level
- Designed the data pipeline from the Amazon Reviews 2023 dataset: matched synthetic personas to real Amazon buyers, calibrated demographics to U.S. BLS Consumer Expenditure Survey data, and constructed 60-product search environments from title-embedding nearest neighbors
- Implemented coordinated LLM-generated fake-review attacks (push, segment-targeted, average-profile imitation) with automated quality checks for length and near-duplicate detection
- Redesigned agent architecture from independent prompts to multi-turn conversations on the Claude API with prompt caching (∼77% cache hit rate), fixed reasoning effort, robust retries, and reproducible run logging
- Ran controlled baseline-vs.-attack experiments across 40 targets and cross-model comparisons across 8 LLMs; observed that attack effects concentrate on targets reaching the first results page, with agents citing fake reviews without suspicion

**Course Research Project, University of Virginia** — Spring 2026
*Privacy Attack via Temporal Embedding Inversion on RAG Systems*

- Proposed a temporal-aware sliding-window method for inverting conversation embeddings leaked from RAG systems, incorporating dialogue history to improve text reconstruction and sensitive-information extraction
- Constructed a 3,000-conversation dataset with three categories of embedded private information (user profile, financial, medical) and evaluated attack success rates across categories

- Analyzed why a non-linear extension (Temporal Context Fusion Network) fails to improve over the linear sliding-window approach, identifying that non-linear projection distorts embedding-space geometry rather than improving alignment
- Achieved best attack success rates of 41% (user profile) and 53% (medical) with the fine-tuned sliding-window variant

**Research Intern, Nanyang Technological University** May 2024 – Sept 2024
*Internal Mechanisms of LLMs in Sentiment Analysis*
Supervisors: Xulang Zhang, Rui Mao, Erik Cambria

- Analyzed hidden states of a fine-tuned RoBERTa model across all 12 layers and 12 attention heads using K-means, spectral, and hierarchical clustering to probe sentiment encoding
- Discovered that only the final three layers encode sentiment-distinguishing representations, and that 3 of 12 heads in layer 11 are redundant (accuracy ∼50% vs. ∼79% for others)
- Demonstrated that hidden states encode part-of-speech information in early-to-mid layers, with performance declining in later layers as representations become more task-specific
- Extended analysis to word sense disambiguation and textual entailment, showing that the model's internal mechanisms are task-driven rather than mirroring human reasoning
- Published at ICDMW 2024

## PUBLICATIONS

**Xue, K.**, Zhang, X., Mao, R., & Cambria, E. "Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis." In *Proceedings of the 2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2024.

**Xue, K.** & Wang, T. "Beyond Contiguous Spans: Evaluating Privacy Filters on Fragmented Private Information." Submitted to *ACM International Conference on Web Search and Data Mining (WSDM)*, 2027. *(Under review)*

## TECHNICAL SKILLS

**Languages:** Python, Java, C, SQL, JavaScript
**Frameworks & Tools:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MATLAB
**Infrastructure:** Git, Linux, LaTeX
**Methods:** Hidden-state probing, Markov chain analysis, clustering (K-means, spectral, hierarchical), benchmark construction, statistical evaluation (bootstrap, permutation tests, cross-validation)

## HONORS & AWARDS

College of Letters & Science Dean's List, University of Wisconsin–Madison